



My doctor is an avatar! The effect of anthropomorphism and emotional receptivity on individuals' intention to use digital-based healthcare services

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ABSTRACT

Alongside teleconsulting platforms, existing digital-based healthcare services propose automated solutions such as medical chatbots to interact with patients. Nevertheless, previous research stress their lack of human-like interactions delaying implementation. The advent of Metaverse may fill this gap by “catapulting” patients and doctors into a parallel virtual reality made up of avatars where interactions are as similar as those in the real world. By leveraging on a sample of 689 participants to an international experimental study, this paper investigates the effect of the human-like interactions (manipulated as low in case of medical chatbots vs high in case of doctors' avatars in the Metaverse) on the individuals' intention to use such digital-based healthcare services. We analyze the relationship through the perceived anthropomorphism mediating the two options. Moreover, by considering the peculiarities of the doctor-patient relationship resulting not only from professionalism but also from social interactions, we analyze the moderating effect of emotional receptivity. Results show that higher level human-like interactions (i.e., doctors' avatars in the Metaverse) positively influences individuals' intention to use such healthcare service via the effect of perceived anthropomorphism. Moreover, such effect is significant only among individuals who exhibit higher level of emotional receptivity.

1. Introduction

The advent of new technologies, such as Augmented Reality (AR), Virtual Reality (VR) and Blockchain, have contributed to the rapid transformation of healthcare business models into digital healthcare business models (Hoppe et al., 2022) in the attempt to fully exploit their disruptive potentialities and guarantee hyper-patient-centric care services. To date, globally, revenue in the digital healthcare newer business models is projected to reach US\$170.20bn in 2023, and it is expected to show an annual growth rate (CAGR 2023–2027) of 10.78 %, resulting in a projected market volume of US\$256.30bn by 2027 (Statista, 2022).

Digital healthcare platforms and solutions (e.g., fully online teleconsulting and platforms connectible with IoT devices for distance telemonitoring) may provide clinicians with a more holistic, continuous monitoring of their patient's health conditions with enhanced opportunities to reach better medical outcomes and improve communication, efficiency, efficacy and the patient overall experience across the care continuum (Su et al., 2021). More importantly, such new digital platforms may bring together ecosystem participants to build new business

models able to co-create goods and services, enable wider patients to reach, access new capabilities, and finally increase business performances (e.g., Schiavone et al., 2021).

The Metaverse is perhaps the latest digital technology employed in the healthcare industry. The Metaverse in healthcare has been valued at USD 6.85 billion in 2021 and it is projected to surpass USD 72.10 billion by 2030 growing at a CAGR of 29.89 % from 2022 to 2030 (Forbes, 2022). As a simulated digital environment that combines multiple elements of existing technologies, such as AR, VR, and Blockchain, along with social media elements, the Metaverse creates spaces that enrich individuals' interaction by mimicking the real world. The result is a parallel world where individuals may virtually live their lives (Lee and Oh, 2022; Sestino et al., 2022a, 2022b; Sestino and D'Angelo, 2023).

By leveraging on these technologies, the Metaverse is rapidly becoming a virtual space in which differences with reality become almost imperceptible, and individuals are made possible to work, meet virtually, do business and purchase goods by acting with their avatars (Seok, 2021). In simple words, an individual's avatar is the virtual equivalent of a single real person. It is a virtual character representing an

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individual in the 3D world (Kato et al., 2022). It is built on the concept of *virtual agents* consisting in a software program that uses scripted rules and, increasingly, AI applications to provide automated service or guidance to humans (Ochs et al., 2010). Such avatars embody high levels of anthropomorphism in terms of attribution of human characteristics to non-human entities, objects, or concepts (Alabed et al., 2022; Guido and Peluso, 2015). In other words, avatars are tailor-made on the basis of the individuals they represent in the real world. Different from virtual agents, avatars act exactly like “puppets” commanded by the real person who controls them (Koike and Loughnan, 2021; Kontogiorgos et al., 2019).

The advent of the Metaverse offers, alongside the many and varied business opportunities, the possibility of accessing a series of services from the comfort of home, such as in the case of healthcare consultation services. This makes the Metaverse the realm of promising solutions in the healthcare and medical field (Sestino and D'Angelo, 2023; Sun et al., 2022). Indeed, the Metaverse in the healthcare services may address specific issues such as the access to care without geographical limitations, opening new channels of treatment, wellness and fitness at lower costs while improving patient outcomes (Thomason, 2021). Especially in the healthcare industry, the Metaverse involves the convergence of three major technological trends, which all have the potential to impact healthcare individually, creating new solutions for care delivery that lower costs and vastly improve patient experiences. These major technological trends refer to (1) telepresence, allowing people to be together virtually, even while they are apart physically; (2) digital twinning, and the aforementioned (3) blockchain, with its ability to let individuals create a distributed internet (Riva & Wiederhold, 2022; Sestino et al., 2022a).

However, by considering the peculiarities of the healthcare and medical field, specifically concerning the relationship between doctors and patients, the Metaverse may help to overcome some critical issues not currently addressed by the current digital business models.

Previous literature acknowledges that the relationship between doctors and patients is built not only on professionalism recognized by the patient to his/her doctor, but it is also identified in a mix of individual characteristics such as empathy, emotionality, positive attitudes for reassuring the patients that may help patients to build a solid relationship with him or her (Goold and Lipkin, 1999; Schmid Mast et al., 2008). This form of interaction in existing online healthcare services, such as in the case of teleconsultation, may remain unaltered as it is technically conducted in an online environment consisting in a remote consultation in which the interactions between a doctor and a patient for diagnostic or therapeutic advice is done through electronic means (O'Cathail et al., 2020; Tortorella et al., 2022).

Other existing digital-based healthcare services, for example teleconsulting delivered through medical chatbots, have been increasingly implemented. These medical chatbots consist of medical AI-powered conversational solutions that may help patients and healthcare providers to easily connect to each other. They capture patient's information and data, assist healthcare professionals based on a set of questions/answers to identify different kind of diseases, disseminate medical information by offering patients' support for self-diagnosis purpose (Divya et al., 2018). Alternatively, they are used to carry on a preliminary screening procedure before to connect patients with real doctors (Bulla et al., 2020). Despite the increasing use of medical chatbots to provide digital-based healthcare services (Dharwadkar and Deshpande, 2018), recent literature highlights the lack of human-like interactions which characterize such service provided through these fully automated medical chatbots (Rosruen and Samanchuen, 2018).

In the case of the Metaverse, many promising opportunities may emerge for digital-based healthcare services as affective communication, communicating with emotion, face-to-face communication and social interactions, that are critical for doctors-patients relationship, may be expressed by the avatars (Córdova González, 2022).

Nevertheless, the translation of these interpersonal relationships into

a virtual environment with an avatar is complex. To date, a recent study (i.e., Numata et al., 2020) focusing on virtual agents' anthropomorphism reports that individuals exhibit positive emotions only when their smile is imitated by the agent's positive expression regardless of their belief (Alabed et al., 2022). The research shows how a positive congruent response by a virtual agent can overcome the effect of believing that the agent is a computer program and thus contribute to achieving affective human-virtual agent interaction. This issue is particularly critical to manage, especially considering the so-called Uncanny Valley theory, which describes individuals' sensation to experiment with a common unsettling feeling people experience when human-like digital tools (e.g., as for humanoid robots, avatars, robots) and audio/visual simulations closely resemble humans in many respects but are not quite convincingly realistic (Berger et al., 2018). Thus, focusing on the different levels of perceived anthropomorphism in designing such new virtual settings may be very important before implementation.

In the attempt to increase attention around the business opportunities of the Metaverse in the healthcare services, this paper builds on the research stream reporting the necessity of maintaining the doctor-patient relationship and interactions also in the virtual world (Lundin and Menkes, 2021). Through an experimental study, the paper investigates the effect of human-like interactions in a digital setting (low for chatbot vs high for the Metaverse-based doctors' avatars) on the individuals' intention to use such digital-based healthcare services. Our research attempts to shed light on the mediating role of the anthropomorphism as perceived by individuals, and on the moderating role of their emotional receptivity (Lee and Lim, 2010; Lennox and Wolfe, 1984). Our results show that the level of human-like interactions (low vs high) positively affects individuals' intention to use a digital-based healthcare service provided by doctors' avatar, through the effect of their perceived anthropomorphism. In other words, the high human-like interaction condition (i.e., such as avatars) exerts a positive effect on individuals' intention to use when they report higher perceived level of anthropomorphism. Furthermore, individuals' emotional receptivity moderates such effect, revealing as significant role only among individuals with a higher emotional receptivity tendency.

2. Theoretical background

2.1. Current advances in digital-based healthcare services: the Metaverse

The Metaverse is currently capturing the attention of researchers, businesses, policymakers, and governments globally. It consists in a medium for communication that bridges physical and virtual experiences. Technically, it results in a digital environment realized by leveraging combinations of AR and VR technologies (Mystakidis, 2022). AR consists in an improved version of the real physical world that is achieved through the use of digital elements, sounds or other sensory stimuli supplied by digital technologies (Carmigniani and Furht, 2011). While AR may be able to control individuals' presence in the real world by using a real-world setting, the VR is completely virtual. This latter requires a headset device, and individuals are controlled by a system to experiment a fictional reality (Jung and tom Dieck, 2018). Indeed, VR consists of a technology capable of replacing reality with a closed simulation resulting in a 3D modeling which requires special viewers connected to a computer or a game console, equipped with powerful graphics processors and other accessories such as inertial sensors, accelerometers, joysticks, and wearable devices (e.g., gloves and visors) making possible to increase individuals' interactivity and immersive sensation (Sherman and Craig, 2003).

Among the new digital technologies, the Metaverse is currently revolutionizing the realm of business opportunities in the medical and healthcare services by increasing patients' chances to access services worldwide (Riva and Wiederhold, 2022). At the industrial level, the Metaverse is not at its first attempt, but it has already numerous

successful cases. For instance, Latus Healthcare in the United Kingdom (<https://latushealth.co.uk/>) has developed a “virtual hospital”, comprising virtual reality hospital environments with virtual agents to assure doctors, nurse and staff a virtual presence online. Such services consist of a digital hospital environment, accessed via a headset, and treatments are focused on counseling and physiotherapy services. Physical therapy utilizes computer vision — for example, the usage of cameras to investigate the variety of movement of broken joints and the improvements patients make during rehabilitation. Another, more complex example is iMining, the first-ever hospital foundation to be launched in the Metaverse, thus directly in the virtual world without having a “twin” in the real life (<https://www.newsfilecorp.com/release/107660/iMining-Technologies-Inc-Enters-the-MetaVerse-and-Becomes-a-Publicly-Listed-Web3.0-Company>).

The emerging literature on the Metaverse in healthcare industries (Riva & Wiederhold, 2022) comprise three fundamental areas of applications that may be summarized as follows: (1) telepresence/treatment; (2) surgical operations, and (3) education. In the case of healthcare services delivered via telepresence, the Metaverse may reduce geographical distances making treatments more accessible even in circumstances of reduced mobility, serious disease, or other circumstances (Werner et al., 2022). Specifically, the Metaverse may allow newer treatments for those patients suffering from chronic diseases by exploiting AR/VR technologies in relaxation and stabilization of their pain. Moreover, the Metaverse may be exploited to provide safe and controlled virtual exposure to traumatic events or to accelerate patient cognitive rehabilitation and recovery through telepresence strategies built on AR/VR technologies (e.g., Petrigna and Musumeci, 2022). Moreover, as shown in the examples above, based on the technologies on which the Metaverse is built, the hospitals may turn in virtual spaces where virtual doctors, staff, and patients co-exist interacting as in the real ones (Dewi and Dharma, 2022).

In the case of surgical operations, the Metaverse may support doctors and medical staff as a digital twin used for remote virtual guidance and support. For instance, by providing navigation solutions to surgeons during surgery, or penetrating patients' anatomy and performing surgery with high accuracy and minimal complications (Thomason, 2021).

Finally, thanks to the opportunity of accessing simulated virtual realities, the Metaverse could facilitate healthcare education through visual, hands-on learning experience for beginners and experienced healthcare professionals improving their understanding of patients' anatomy and how to perform complex surgical operations (Kye et al., 2021; Petrigna and Musumeci, 2022).

From a final user's perspective (i.e., the patients), the confluence of the aforementioned new technologies activating the Metaverse may enable healthcare specialists to deliver a suite of highly integrated, intentional, and individualized care, unfettered by the siloed nature of existing healthcare models in a “parallel” daily life environment (Sun et al., 2022). Given the advent of the new technologies (e.g., AI, IoT, and so on) and their combined use healthcare and medical environments and services could be fully transformed (Bayramzadeh and Aghaei, 2021). This may require reimagining the doctor-patient relationship (Córdova González, 2022). The Metaverse may foster both doctor-patient and doctor-doctor communication enabling unprecedented levels of sophistication in prevention, diagnosis, and treatment, by conducting the experience to a next-level. However, despite the valuable opportunities, and by considering the complexity of the doctor-patient experience in a virtual world made by virtual agents, the effects on the final users' intention to use are still overlooked.

2.2. Human-like agents and individuals' intention to use

The individuals' intention to use digital-based services delivered by human-like agents in the form of virtual agents (such as fully automated and computer-driven chatbots) or intelligent agents (such as partially automated and human-driven avatars) has attracted scholars' attention.

The following table sheds light on the extant literature reporting the main elements that may influence individuals' intention to use services delivered by human-like agents not just limited to the healthcare industry (Table 1).

Virtual agents (or intelligent virtual agents, IVAs) consist in a contact center capability using a virtual character, generally created through computer generation, animation, and AI techniques in the attempt to support the individuals in the digital environments (Holz et al., 2009). In simple words, such agents are software programs that, by using scripted rules and automated AI-based functionalities, provide automated service or guidance to humans. Thus, virtual agents may appear as a chatbot i.e., agents simulating a human conversation in a digital environment, or as an avatar i.e., a digital reproduction of a human in the digital world (Norouzi et al., 2019). From a functionality perspective, while a chatbot generally is a digital tool that simulates human conversation through auditory or textual methods, an avatar is still controlled by the human behind it. This makes interactions with an avatar more “humanized” as it performs as a real and actual assistant, able to answer specific questions, perform specific tasks, and make recommendations. Thus, an avatar is a digital character, powered by AI, that lives in a virtual setting, like a game, social network, dating app, or online world. The literature traditionally distinguishes these two types of virtual agents. While avatars are controlled by humans appearing as a “virtual edition” in the virtual world of the human controlling them, the chatbots are controlled by a computer (Shaked, 2017). Hence, the interaction with an avatar qualifies as a computer-mediated communication, whereas the interaction with a chatbot qualifies as a human-computer interaction (Morkes et al., 1999). Thus, an avatar represents a virtual equivalent of a single and real individual translated into the virtual world, built on AI and software design, partially automated in some movements and actions, but responding and acting mainly as its user (Sestino et al., 2022b). By considering the healthcare sector, similar to other delivery services, the interaction in the Metaverse is allowed by a “mixed” tool made of virtual agent-based avatars. “Such avatars can be personalized, and the cultural, physical, and social characteristics are different from reality, but the avatar can communicate with other entities and achieves goals” (Petrigna and Musumeci, 2022, p. 64). This may result in the creation of a virtual environment with three-dimensional spaces and living avatars in which, differently from teleconsulting simply using a VR headset, individuals' may meet their doctor through the device as a digital avatar to have their consultation held in a virtual space. Moreover, such virtual spaces may allow them to meet up with fellow patient communities to share, in their avatar form, their treatment progress, discuss new developments pertinent to their treatments, and so on (Tan et al., 2022; Thomason, 2021).

Despite the great opportunities, similar to the virtual agents, the translation into the online environment of characters representing real individuals, or the proposition of human-like agents able to act as human beings, has deeply fascinated and interested researchers during the last years (Weitz et al., 2021) in the attempt to replicate characteristics, behaviors, actions, emotions, visual appearance typically ascribable to humans (Pelachaud, 2009).

On the basis of the extant literature, the role of real individual similarity (or identification) to their avatar or to the online available human-like agents, (e.g., Jahn et al., 2021; Keeling and McGoldrick, 2008; Kim and Kim, 2016; Yang et al., 2021), along with the emotional components of digital interactions such as empathy and social interaction (Hall et al., 2005; Cordar et al., 2014; De Rosi et al., 2005; Jahn et al., 2021) with human-like intelligent agents such as the avatars seems to be important in influencing individuals' intention to use such digital service.

Thus, in the healthcare services where the relationship between doctors and patients is built not only on the professionalism of the doctors, but also on emotional aspects (Goold and Lipkin, 1999; Schmid Mast et al., 2008), the latter characteristics may assume a crucial role in the patients' intention to use doctor-avatar in the Metaverse rather than

Table 1

Main variable influencing individuals' intention to use human-like agents.

Source	Setting	Method	Involved variables	Main arguments and findings
Cordar et al. (2014)	Virtual human avatars, interacting with medical students.	Quantitative, experiment; Adults: The study has been conducted among a sample of medical students.	-Empathy -Background story -Intention to use	"Medical students who interacted with a virtual human with a back story, when interacting with a standardized patient, were perceived by the standardized patient as more empathetic compared to the students who interacted with the virtual human without a back story" (p. 144).
Hall et al. (2005)	Virtual bots for anti-bullying education purpose interacting with children.	Quantitative, experiment; Children: The study has been conducted among a sample of children aged 9–12.	-Facial expression -Tone of voice -Empathy modulation through the identification with the victim's character	-"The interaction between characters is the driving force for the development of narrative and the whole agent architecture and the surrounding framework allowing agents to interact with each other have to be completely in place before any real testing of the narrative" (p. 12) produced can be carried out. -“When the animation that represents it in the graphical world must be visually correct, and this requires the character to position itself so that this is true” (p. 13) -The tones of voice (and related expressions) must be appropriate for the audience (i.e., adults for adults, younger for young individuals). To being used and being effective, the virtual agents should be able to be emphatic, capable of advice-giving, e. g., “also leveraging on the ability to recognize the degree of involvement of the user and to manifest reciprocity of social relationship seems to be more important than displaying realistic facial expressions of emotions” (p. 71). Individuals' gender identification may affect their intention to use avatars: Results have shown significant differences in intimacy when a female user had the same gender avatar. However, there was no significant difference in intention of use.
De Rosi et al., 2005	Human-like virtual bots in wellness	Quantitative; experiment; Young adults: The study has been conducted among teenagers seeking for advice in a consultancy (nutrition) setting, as a part of iterative prototyping of health promotion dialog	-Level of the involvement -Degree of the initiative -Emotions in facial expressions (e.g., clearness; plausibility; reasonableness, persuasiveness, naturality, and so on) -Empathy	The different forms of feedback and avatar design influence individuals' intention to use avatar-based services, by satisfying the three psychological needs. Moreover, the autonomy for decision freedom is negatively related to reuse intention.
Yang et al., 2021	Human-like artificial agents as avatar	Quantitative, experiment; Young: The study has been conducted among a sample university students	-Gender -Identification with the avatar -Empathy -Intention to use -Avatar perceived reality	
Jahn et al., 2021	Human-like intelligent avatar in food service	Quantitative, experiment; German Adults: randomly selected participants	-Gender -Identification with the avatar -Avatar similarity (high similarity = same gender; low similarity = different gender) -Gamification: Embodied feedback (The gamification design element embodied feedback has been operationalized by providing users feedback about their performance through the body shape of their avatar) -Need for autonomy -Social Relatedness -Intention to use	
Wang et al., 2020	Human-like avatar used to support consumers' fitness activities and healthy food consumption, to have a sustainable healthy lifestyle	Quantitative, experiment; German Adults: randomly selected participants	-Feeling -Absorption -Identification -Immersive experience -Enjoyment -Self-presence -Intention to use	Avatar-related immersive experience, self-presence and enjoyment positively impact individuals' intention to use
Keeling and McGoldrick, 2008	Human-like avatar to support consumers on a retail fictitious store (website)	Quantitative, experiment; Young: The study has been conducted among a sample university students	-Identification (similarity with the avatar) -Friendliness -Social competence -Social interaction -Trust -Intention to use	Despite results indicating an advantage of attractiveness for individuals' rated sociability, similarity, and relationship quality over time, such elements do not affect the effects of avatars' attractiveness on trust or usage intentions. However, individuals' perceived avatar attractiveness may neutralize their negative intention to use.

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Table 1 (continued)

Source	Setting	Method	Involved variables	Main arguments and findings
Kim and Kim, 2016	Human-like avatar in an immersive 3D virtual environment (for alcohol-impaired driving monitoring purpose)	Quantitative, experiment; Adults: Volunteer randomly selected participants	-Identification -Physical attractiveness -Emotions (i.e., control) -Intention to use	Results show that having an individualized avatar may enhance the emotional tie between the creator and the avatar. Individuals' option to control their avatars, takes greater responsibility for what happened to the avatar.

chatbots, since avatars are perceived as more human-like interactive.

2.3. The role of perceived anthropomorphism

The concept of individuals' perceived similarity to human-like agents has been debated in the literature (e.g., Jahn et al., 2021; Yang et al., 2021) and it has often been operationalized through the construct of individuals' perceived anthropomorphism (Blut et al., 2021). Coherently, previous research investigates the role of individuals' perceived anthropomorphism on non-physical objects such as personal computers (Waytz et al., 2010), non-human agents (Epley et al., 2008), and recently, human-like intelligent agents such as the avatars (Banks and Bowman, 2016; Lugin et al., 2015; Sestino and D'Angelo, 2023). By considering individuals' perceived anthropomorphism, previous research also shows that, according to the so-called Uncanny Valley theory, such perceptions of robotized digital tools as anthropomorphic may boost psychological warmth but reduce attitudes owing to uncouthness. Competency evaluations are far less impacted and unaffected by a change in attitudes (Kim et al., 2019). Moreover, in several research settings, despite the platform's overall pleasant vibe, the human-like virtual influencer gets far less favorable feedback, which supports the Uncanny Valley theory (Arsenyan and Mirowska, 2021). However, as discussed below in relation to virtual agents' exploitation, counterintuitively in the healthcare field such proximity solutions could instead be better accepted. Anthropomorphism can become a crucial and positive factor when it comes to health (for example, because avatars are moved by humans, and humans may be comfortable in other humans in helping them), but it is avoided when machines have different purposes (robots of any other type/setting).

To the best of our knowledge only few studies have investigated the role of individuals' perceived anthropomorphism comparing virtual agents (e.g., Banks and Bowman, 2016; Blut et al., 2021). Moreover, no studies have investigated such topic in the healthcare sector considering together the levels of human-like interactions and the role of end-users-related individual differences.

The construct of anthropomorphism refers to the individuals' tendency to perceive inanimate objects as human-like entities (Guthrie, 1993). It consists in the inference process of attributing external characteristics, motivations, behaviors, and underlying states that are typical of humans to objects and non-human agents (Epley et al., 2007; Guido & Peluso, 2015). Individuals' perceived anthropomorphism may directly influence their intention to use a human-like intelligent agent, both depending on the agent type e.g., robot gender, individual similarity, and so on, or the type of service (Blut et al., 2021; Sestino and D'Angelo, 2023). This individuals' intention is mainly driven by the similarity they attribute to the digital tools, such as chatbots (Sheehan et al., 2020) or avatars (Jahn et al., 2021).

In the healthcare sector, despite the efforts of medical chatbots to reach increased level of anthropomorphism (Bulla et al., 2020), the avatars may be better welcomed by final users because they are perceived more similar to the humans (e.g., doctors or clinicians) they represent in the virtual world. Moreover, since the avatars are not automated by computers but controlled by humans, they may embody all the positive sides, acting like humans, and not incurring in technological errors because there are humans behind them. Furthermore, fully-automated virtual agents such as chatbot are exposed to errors due

to the limited knowledge base on which such agents are built, negatively influencing individuals' intention to use such tools (Sheehan et al., 2020).

The transmission of emotions is even more complex in digital environments, and it may be particularly crucial especially in the healthcare sector. The importance of affective communication during doctors-patients' interactions cannot be neglected, even in the real world (Córdova González, 2022). Thus, the mix of empathy, emotionality, positive and reassuring attitudes required from patients (Goold and Lipkin, 1999; Schmid Mast et al., 2008) cannot be underestimated when moving the doctor-patient relationships in the digital environment. They may play a crucial role in the patients' intention to use.

2.4. The role of emotional receptivity

Previous literature recognizes the importance of customers' emotions related to non-verbal communication qualifying them as both antecedents of emotional manifestations (Diefendorff and Richard, 2003; Diefendorff et al., 2011) and predictors of satisfaction deriving from using a specific service (Delpechitre et al., 2018; Lamont and Lundstrom, 1977). This latter refers to the concept of emotional receptivity. The emotional receptivity indicates a trait, related to individuals' preference to physically express positive emotions in communication, especially by focusing on others' non-verbal expression of emotions, such as gestures, body or facial expressions, variations in the interlocutors' voice, tone, pitch, loudness, and displayed empathy (Lee and Lim, 2010; Lennox and Wolfe, 1984). Lee and Lim (2010) defined individuals' emotional receptivity as a "disposition toward experiencing a preferred level of emotional intensity" (p. 1151). The relevance of non-verbal communication and emotional receptivity is central in Mehrabian's seminal article (1981) proposing the so-called 55-38-7 model and explaining that the 55 % of the communicative message is deduced through body language (e.g., as for gestures, facial expressions, postures); 38 % is deduced from the paraverbal aspects (e.g., as for tone, rhythm, timbre of the voice); and 7 % is inferred from the words spoken, i.e. from the verbal content. Accordingly, facial and body expressions may play a central role in expressing human emotions. They can even replace verbal communication in terms of "closeness", empathy and understanding, allowing judgments to be made on intelligence, personality, and mood (Ekman et al., 1988). Despite facial and body expressions, gestures seem to represent only an alternative to verbal communication. Nevertheless, such actions are an essential part of the multichannel communication system, offering the interlocutors different options through which convey the message (Kenny, 2018; Tannen and Saville-Troike, 1985).

In the healthcare sector, the doctors' non-verbal communication during medical visits plays an important role in assessing patient's compliance and satisfaction with healthcare treatments (Goold and Lipkin, 1999; Ong et al., 1995). Indeed, doctors' medical profession requires they are both clinically competent and empathetic towards the patients (Kerasidou and Horn, 2016). Both verbal and non-verbal communication are important skills for the development of a relationship of trust between patients and doctors or other health professionals. Previous studies show that the doctors' non-verbal emotional communications generate positive empathy. This has positive effects on patients' perception of the overall quality of the healthcare service

(Paternotte et al., 2015; Ranjan et al., 2015). Some doctors' behaviors while approaching the patient, such as physical proximity, leaning forward, nodding head and sitting closer have been associated with patients' satisfaction. Patients are receptive these non-verbal approaches and are often looking for a "total guidance", both in professional and emotional terms (Hall et al., 1995). Accordingly, the literature reports how socio-emotional interchange (e.g., approvals, laughter, legitimacy, empathy) largely influence how patients feel after the consultation (Pawlikowska et al., 2012). Non-verbal communication, as a fundamental factor of such socio-emotional exchange (Roter et al., 2006), has been recognized as a stimulus for positive emotions. This kind of communication is essential for patients' perceived quality of the healthcare service (Laine et al., 1996) and it influences the overall satisfaction of the delivered service.

Previous studies largely confirmed that the socio-emotional dimensions of the doctor-patient relationship depends in a moderated manner on the doctors' ability to understand the patient's non-verbal cues as well as his/her ability to intentionally communicate through non-verbal actions (DiMatteo and Taranta, 1979). However, despite the way the doctors behave non-verbally affects patient outcomes, such as patients' intention to use a consulting service and their level of satisfaction, the doctors' non-verbal behaviors may also depend on doctors' personal attributes (Mast, 2007), that magnify patients' emotional states looking for "a welcoming approach" (Clark et al., 2003).

Thus, based on above, we hypothesize that:

Hypothesis. The level of human-like interaction (low vs high) increases individuals' intention to use digital-based healthcare services, via their perceived anthropomorphism. However, this is moderated by individuals' emotional receptivity.

Fig. 1 below presents the proposed conceptual framework.

3. Materials and methods

3.1. Procedure and questionnaire

To test the working hypothesis, an experimental design based on a two-cell experiment has been conducted by manipulating the level of human-like interactions in digital-based healthcare services.

The two-cell experiment built on two scenarios has been conducted among a sample of 689 international participants, randomly recruited online through M-Turk platform (Aguinis et al., 2021). Half participants have been exposed to a scenario in which a target patient had the possibility of using a remote teleconsultation in an online environment through the use of a chatbot able to interact with the patient asking him/her about their symptoms, allergies, and so on in a private chat. This setting allowed us to manipulate the low human-like interaction condition. The other half of participants has been exposed to a scenario where a patient had the opportunity to enter the Metaverse and interact with his/her GP for the desired teleconsultation. In this setting doctors were faithfully reproduced in an online digital environment where movements, expressions, gestures by an avatar allowed us to reproduce a context of high human-like interactions. The two conditions were built based on the literature (Table 1). A pre-test was performed among a sample of 40 on-line participants who rated the perceived level of human-like interaction about the chatbot vs the avatar in the Metaverse

(1 = "Totally non-human-like interactive", 7 = "Totally human-like interactive"). Results have shown that the average score of human-like interaction level assigned to higher condition ($M = 2.656$, $SD = 2.378$) vs low condition ($M = 2.042$, $SD = 1.322$) was not significantly different ($t = 1.00$, ns). Thus, any possible differences resulting by the study should be not attributed to non-human-like interaction perceptions.

The administered questionnaire was built through the Qualtrics platform and it comprised four sections. First, we asked participants to rate their emotional receptivity on a five-item scale based on Lee and Lim (2010) (e.g., "It is necessary to show too much facial expressions"; $\alpha = 0.883$), assessed on a seven-point Likert scales (1 = "completely disagree", 7 = "completely agree"). In the second section, participants were randomly assigned to two different scenarios/settings within a two-cell experiment that manipulated the level human-like interaction in the two digital-based healthcare services (low level, by using chatbot vs high level, by using avatars). Then, in the third section, we asked for participants' intention to use a digital-based healthcare service by using two items based on Fishbein and Ajzen (1977) (e.g., "I intend to use such teleconsulting modality in the future"; $\alpha = 0.901$), and perceived anthropomorphism assessed by using four items drawn from McLean and Osei-Frimpong (2019) (e.g., "The interaction experience with the doctor avatar is close to that with a human being"; $\alpha = 0.892$). Both individuals' intention to use and perceived anthropomorphism were assessed on a seven-point Likert scales (1 = "completely disagree", 7 = "completely agree"). Finally, in the fourth section, individuals indicated their socio-demographic data (i.e., age, education, nationality, employment). Furthermore, based on Oppenheimer et al. (2009), we used in the questionnaire an attention-check item ("If you are reading this question, please select answer 5") to assure respondents read instructions and questions carefully during their duty. This item allowed us to detect and remove the aforementioned 33 respondents who negligently answered the questionnaire, avoiding noise in the data.

3.2. Sample

As reported above, the survey was built with Qualtrics and administered randomly for a period of 8 weeks (June 1st 2022–August 31st 2022), always accessible 24 h, through the platform M-Turk (Aguinis et al., 2021). Participants accessed the questionnaire and, after reading the information set assuring anonymity and confidentiality, completed the requested measures described above. Despite the original sample contained 733 questionnaires, we removed the 33 participants who failed the attention check question, and additional 11 incomplete questionnaires. Thus, the final sample consisted in 689 participants aged between 19 and 78 ($M_{age} = 36.68$; $SD_{age} = 12.11$). The sample included 278 male (54.86 %) and 311 (45.11 %) female participants. One-hundred eleven (16.11 %) declared to have a high school diploma, or lower than high school diploma; five-hundred fifty-five declared to have a degree (respectively 243 participants held a B.Sc.; 312 participants held a M.Sc.) and only 23 gained a Ph.D. degree (3.34 %). A detailed table showing the total sample composition is reported below where distribution of the sample is shown also by level of employment and geographical region (Table 2).

Following Hair et al.'s (2003) criterion, we use at least 5 (ideally 15–20) observations for each independent variable included in the

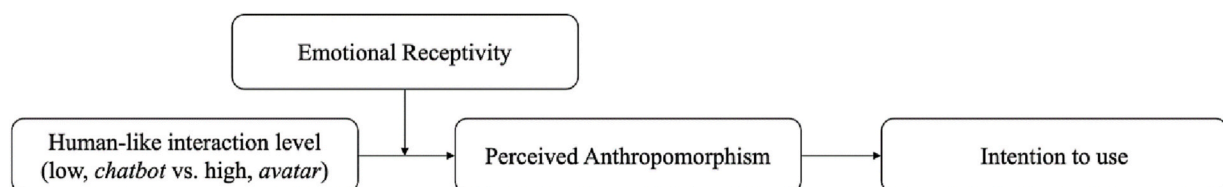


Fig. 1. Perceived anthropomorphism and emotional receptivity as antecedents of individuals' intention to use digital-based healthcare services.

Table 2

Descriptive sample analysis (gender, education, employment, geographical region).

Gender		
Male	378	54.86 %
Female	311	45.14 %
Education		
High school or lower	111	16.11 %
B.Sc.	243	35.27 %
M.Sc.	312	45.28 %
Ph.D. or superior	23	3.34 %
Employment		
Students	241	34.98 %
Employed	325	47.17 %
Not employed	112	16.26 %
Retired	11	1.60 %
Geographical region		
Europe	219	31.79 %
Northern America	236	34.25 %
Southern America	88	12.77 %
Asia	123	17.85 %
Africa	23	3.34 %

regression equation. The final collected sample has been considered adequate to test the proposed hypothesis, involving a dependent variable and a total of four independent variables. Specifically, the predictor (i.e., Human-like interactions level), the moderating variable (i.e., Emotional Receptivity), the interaction term (i.e., Human-like interaction level $\times$ Emotional Receptivity), and finally, the mediator (i.e., Perceived Anthropomorphism).

4. Results

The analysis has been performed by using IBM SPSS 26. The PROCESS macro (Model 7; Hayes, 2018) has been launched to sage the effects derived by the proposed moderated-mediation model. Results have been evaluated in the following three steps. First, individuals' perceived anthropomorphism (continuous variable) has been expressed as a function of the human-like interactions level (as a continuous), individuals' emotional receptivity (continuous, mean-centered variable), which operated as the moderator, and their interaction. Second, individuals' intention to use digital-based healthcare service (continuous variable) has been expressed as a function of their perceived anthropomorphism, which served as the mediator, together with controlling for the direct effect of the human-like interaction level. Finally, in the third step, the indirect effects have been evaluated. Specifically, the indirect effect of the human-like interactions level on the intention to use digital-based healthcare service, through the effect of perceived anthropomorphism related to the different levels of individuals' emotional receptivity (Table 3).

The statistical analysis carried out returned that both the human-like interactions level ($b = 0.581$, $p < 0.001$) and individuals' emotional receptivity ($b = 0.677$, $p < 0.001$) are positively related to individuals' perceived anthropomorphism. Furthermore, results show the effect of the interaction term (human-like interaction level $\times$ emotional receptivity) is positive and significant ($b = 0.312$, $p < 0.001$). In turn, the perceived anthropomorphism is also positively related to individuals' intention to use digital-based healthcare service ($b = 0.681$, $p < 0.001$), even when controlling for the direct effect of individuals' perceived human-like interaction level ($b = 0.704$, $p < 0.001$).

Table 3

Results of the moderated-mediation analysis.

Step 1.	<i>b</i>	<i>SE</i>	<i>t</i>	<i>p</i>
Dependent variable. Perceived virtual agent (doctors' avatar) anthropomorphism (<i>Me</i>)				
Constant	4.509	0.096	46.969	0.000
Human-like interaction level (<i>X</i>)	0.582	0.120	4.850	0.000
Emotional receptivity (<i>Mo</i>)	0.588	0.125	4.704	0.000
<i>X</i> $\times$ <i>Mo</i>	0.334	0.065	5.138	0.000
$R^2 = 0.438$, $MSE = 1.904$, $F(4, 678) = 60.248$ $p = 0.000$				
Step 2.	<i>b</i>	<i>SE</i>	<i>t</i>	<i>p</i>
Dependent variable. Intention to use (<i>Y</i>)				
Constant	2.911	2.302	1.265	0.000
Human-like interaction level (<i>X</i>)	0.518	0.089	5.820	0.000
Perceived virtual agent anthropomorphism (<i>Me</i>)	0.640	0.066	9.697	0.000
$R^2 = 0.8914$, $MSE = 1.584$, $F(3, 678) = 102.675$, $p = 0.000$				
Step 3.	<i>b</i>	<i>SE</i>	<i>LLCI</i>	<i>ULCI</i>
Conditional indirect effects of <i>X</i> on <i>Y</i> at low and high levels of <i>Mo</i>				
Low level of <i>Mo</i> ($M - 1SD$)	0.143	0.083	-0.017	0.306
High level <i>Mo</i> ($M + 1SD$)	0.488	0.106	0.293	0.703
Index of moderated mediation	0.182	0.045	0.104	0.281

$N = 689$; X = predictor (continuous and mean-centered); Mo = moderator (continuous and mean-centered); Me = mediator (continuous); Y = ultimate dependent variable (continuous); $LLCI$ = lower-level confidence interval; $ULCI$ = upper-level confidence interval.

Furthermore, on the basis of the moderated-mediation model, results also revealed that the interaction term exerted an indirect effect on to use digital-based healthcare service (index of moderated mediation: $b = 0.166$, 95 % Confidence Interval [CI]: 0.106, 0.287). Moreover, consistently with the proposed hypothesis, the human-like interaction level exerted an indirect effect on intention to use digital-based healthcare service (through the effect of perceived anthropomorphism), resulting positive and significant among participants with higher level of emotional receptivity ($M + 1SD$: $b = 0.498$, 95 % CI = 0.300, 0.719) and non-significant among those with a lower level of emotional receptivity ($M-1SD$: $b = 0.126$, 95 % CI included zero).

5. Discussion and implications

5.1. Overall discussion

The technologies behind the Metaverse are applied to translate physical actions into the virtual sphere (Haenlein and Kaplan, 2019; Kaplan and Haenlein, 2020). These new technologies have seen significant advances over the recent years (Kim, 2021). VR, AR, AI, human-computer interaction, blockchains, cloud computing and mobile networks, are currently driving the development of the Metaverse (Lee et al., 2021). Today, even users seem to be more prepared to accept virtual worlds. This is due, on one side, on the greater maturity of the underlining technological infrastructure and the development of complementary technologies that allow users to be entirely immersed in virtual worlds by wearing headsets, gloves, or entire bodysuits instead of merely glancing at their computer screens or interacting with a fully automated chatbot. On the other side, the pandemic COVID forcing social distancing enhanced a wider use of mobile technologies that paved the way to the recent interest in the Metaverse. New business models are emerging around the world based on the new technologies of the Metaverse and many business opportunities are arising in various

sectors disrupting the way we interact and bringing notable implications for businesses and society (Dwivedi et al., 2022; Polyviou and Pappas, 2022).

In the healthcare and medical field, the Metaverse represents the opportunity to reshape the world of digital-based healthcare services (Sun et al., 2022). The advent of the Metaverse technologies and their amalgamation may address specific issues such as the patients' access and experience opening up to new ways of treatments lowering costs while improving patients' outcomes by ensuring immersive, intimate and personalized patients' care (Chengoden et al., 2022; Liu et al., 2022; Thomason, 2021).

Nevertheless, the development of new business models for digital-based healthcare services built around the Metaverse technologies cannot leave aside the key relationship between doctors and patients that characterize any healthcare service. By "catapulting" patients and doctors into a parallel virtual reality made up of avatars where interactions are as similar as possible to those in the real world, a mix of individual characteristics need to be considered when building a doctors' avatar, beside their professionalism. Empathy, emotionality, positive attitudes capable of reassuring the patients and help them to build a solid relationship (Goold and Lipkin, 1999; Schmid Mast et al., 2008) with the doctor's avatar are of key importance. They should help to prepare patients to move into virtual worlds where social interactions with practitioners and health professionals cannot be underestimated to promote access to this kind of new digital-based healthcare services.

This study attempts to provide a response to the plea by Dwivedi et al. (2022) for research examining how the Metaverse can contribute to telemedicine and other healthcare activities. Specifically, this paper focuses on patients' intention to use a digital-based healthcare service provided by a doctor's avatar in the Metaverse as opposed to a medical chatbot. Previous studies acknowledge that, while medical chatbots may be associated to low level of human-like interactions due to their full dependance to fully-automated AI-powered conversational solutions, the avatars in the Metaverse may be associated to high level of human-like interactions because a sort of individuals' extended-self (Sestino et al., 2022b).

5.2. Theoretical implications

Building a manipulation of human-like interactions based on previous literature, our research provides both theoretical and managerial implications.

From a theoretical perspective, we advance previous research concerning the field of technologies' intention to use. Specifically, we contribute to the literature on individuals' intention to use digital-based healthcare services (Yang et al., 2021; Keeling and McGoldrick, 2008; Kim and Kim, 2016; Jahn et al., 2021) by demonstrating that in newer virtual settings, both service-related characteristics (i.e., human-like interactivity, anthropomorphism), and consumers-related individual differences (i.e., emotional receptivity) influence individuals' intention to use. Moreover, our research contributes to the field of study related to the digital-based healthcare services and the humans-to-machine interactions. Specifically, we show how different levels of the human-like interactions effect individuals' intention to use via perceived anthropomorphism (Blut et al., 2021; Jahn et al., 2021). Our experimental study, conducted in the field of digital healthcare services, highlights important implications in terms of the anthropomorphism of virtual agents, towards which individuals (patients) perceive some traits of humanity (Alabed et al., 2022). Furthermore, our study contributes to the research stream related to doctor-patient communication approaches (Ong et al., 1995) and the role of emotions in these kind of digital-based relationships (e.g., as in Córdova González, 2022). We suggest that individuals' intention to use new digital-based healthcare services not only depends on the efforts regarding the replication of human-related features (i.e., human-like interactions, anthropomorphism), but it may require a higher level of emotionality receptivity.

Moreover, we add knowledge on the research stream of research related to individuals' negative reactions to anthropomorphic digital tools. Indeed, despite the so-called Uncanny Valley perspective, some humanized virtual or digital tools may engender some negative reactions in final users, because they could be uncanny when seeing virtual agents that behave acting as humans (e.g., Berger et al., 2018). In our research, we conversely confirm that higher perceived anthropomorphism led to a greater intention to use. This may involve that the humanization of a virtual agent - by acting on its anthropomorphism - may help consumers in being more comfortable when using the service, especially in the healthcare sector where, by considering the relationship of close trust and human contact that is established between doctor and patient, savoring a total transposition of the same figure in which the patient has extreme trust in a virtual context can be particularly appreciated.

5.3. Managerial implications

From a managerial and service-design perspectives, our findings provide important insights for understating individuals' reactions towards digital-based healthcare services and how to design such new services to make them more effective to push individuals' intention to use. Digital-based healthcare services should be designed increasing the level of anthropomorphism of the digital interlocutors (either in the case of chatbots or avatars). Moreover, design efforts should be directed to increase the ability of virtual interlocutors to satisfy certain levels of individuals' emotional receptivity. Thus, new digital-based healthcare services should be equipped with anthropomorphism characteristics and ability to have non-verbal communication skills.

Our study shows how the interactions with doctor avatars compared to automatic medical chatbots allow patients to create a more empathetic and therefore more effective relationship with the doctor. The simulation of dialog between people, even if not real, inspires less fear, predisposing to greater openness in communication compared to the answer to "cold" questionnaires administered via chat. The possibility of communicating with a virtual avatar with a defined identity and answering questions about their experience by interacting with a virtual avatar of a real doctor makes patients feel more at ease allowing people to be more inclined to communicate for an effective use of the service. Furthermore, we suggest how the construction of fully-virtual environments, should not only not only resemble the real world as possible, but it should be able to allow to experience similar emotions, sensations, and experiences. Moreover, by considering the typicality of a Metaverse-based healthcare service delivery, which does not require minimal human interaction, but is totally digital native, we suggest that the Metaverse may represent one of the most important technological advance able to boost new digital business models in healthcare.

By focusing on the Metaverse-related opportunities, the most important ones may be directed to the creation of virtual hospitals providing a wide range of healthcare services, as already happening in the Arab Emirates where the first hospital in the Metaverse is opening avenues for future digital structures qualifiable as a sort of "meta-hospital" (Forbes, 2022). Thus, these new business models may be able to offer totally restructured health services because of the technologies on which they are built as for teleconsulting, to improve the quality of visits and remote consultations; for the maximum protection of personal data; for clinical telerehabilitation (e.g., through the involvement of patients and the effectiveness of the rehabilitation path, also overcoming the monotony of many rehabilitation sessions as well); for health and mental well-being e.g., by offering the possibility of interacting in an alternative and recreated digital context and suitable for the existing pathologies; for clinical collaboration, by changing some aspects from telepresence, and ranging from simple multidisciplinary consultation to remotized collaborative surgery.

6. Conclusions and limitations

The landscape of digital-based healthcare services is being reshaped by current technology breakthroughs. Existing digital business models provide automated services to connect with patients, such as medical chatbots. Despite this, past study has shown their lack of human-like interactions. In this scenario, the introduction of Metaverse may fill this void by “catapulting” patients and clinicians into a parallel virtual reality made up of avatars, with interactions as close to those in the real world as feasible.

The opportunities enabled by new technologies and the transition of traditional business models to digital business models are disruptive in the healthcare sector. Based on previous literature we demonstrate that, by considering the currently known virtual agents characterized by different levels of human-like interaction because of their technicalities (i.e., low interaction levels available for chatbots, and high interaction levels available for human-like avatars in the Metaverse), individuals’ reaction may change. More specifically, in this paper we demonstrate that the human-like interaction level (low vs high) may positively influence individuals’ intentions to use through the effect of their perceived anthropomorphism. Moreover, by considering the peculiarities of the doctor-patient relationship resulting not only from professionalism, but also from emotionality, empathy, and non-verbal communication, we postulate that such an effect may be magnified by patients’ emotional receptivity seeking, especially in fully-virtual environments.

Despite the potential important implications of the research, our study presents some limitations that may open avenues for future work. Firstly, we only looked at individuals’ emotional receptivity to express the need for non-verbal communication in a doctors-patient

relationship. Thus, future studies could investigate other individual differences such as individuals’ personality, socio-demographic (i.e., the role of age), or other doctors-patients relationship-related features such as trust. Future research efforts may also try to expand our proposed conceptual model by including other potentially relevant variables such as those related to individuals’ emotional states. Moreover, as we manipulate the human-like interaction level by using medical chatbots (low level) vs Metaverse-based avatars (high level), future studies could focus on different digital-based healthcare services to expand our findings. Finally, future studies may also consider different kinds of healthcare services (e.g., teleconsulting, rehabilitation) to investigate how to co-design effective virtual environments, so to further shed light on possible differences in patients’ and doctors’ reactions. From a managerial perspective, future studies may try to focus on how to manage this complex digital transition, and the related impacts on healthcare business performances.

The Metaverse presents the potentials to facilitate healthcare as a service through the concept of the avatar that can offer an innovative element for the benefit of both patient and healthcare professionals (Forbes, 2022), becoming a real concrete example of “technologies serving the humanity” (Kotler et al., 2022).

We hope with this research to offer some contributions to the nascent literature on the individuals’ intention to use such digital-based healthcare services. Nevertheless, many important questions remain regarding equal access to the Metaverse technologies, and the barriers to new healthcare services based on income levels (Dwivedi et al., 2022).

Data availability

The data that has been used is confidential.

Appendix A

Emotional receptivity (Moderator, Mo)

Source: Lee, Y. H., & Lim, E. A. C. (2010). When good cheer goes unrequited: How emotional receptivity affects evaluation of expressed emotion. *Journal of Marketing Research*, 47(6), 1151-1161.

Items:

- The use of hands and other body movements is very helpful in facilitating communication
- It is necessary to show too much facial expressions
- People should show a lot of facial expressions when talking
- I pay close attention to people’s gestures when talking to them
- When communicating, a person should use a lot of variation in his/her voice tone, pitch, and loudness

Level of Human-like interaction	
Low (Chatbot-based)	High (Avatar-based)
Imagine having a severe headache and tenths of a fever and wanting to go to your doctor to ask for a diagnosis and get a prescription for the most suitable therapy. You can't move from your home, but you have the possibility to access an online platform , made available by the medical clinic you trust, to run a consulting with a doctor chatbot . Once you have accessed the platform, the answering doctor bot it's ready to chat with you, asking for your symptoms, pains, allergies to particular medicines, and so on, in the attempt to diagnose your symptoms and suggest the better cure. The doctor then explains to you that the illness is simply due to the change of season, finally suggesting the most suitable medicines, inviting you to take care of yourself.	Imagine having a severe headache and tenths of a fever and wanting to go to your doctor to ask for a diagnosis and get a prescription for the most suitable therapy. You can't move from your home, but you have the possibility to access to the metaverse version of the medical clinic you trust, to interact with the avatar of your real doctor . Once you have accessed the platform, the answering avatars' doctor is pretty similar to the original one in its appearance , and it's ready to talk with you, asking for your symptoms, pains, allergies to particular medicines, and so on, in the attempt to diagnose your symptoms and suggest the better cure. The doctor then explains to you that the illness is simply due to the change of season, finally suggesting the most suitable medicines, inviting you to take care of yourself.

Perceived Anthropomorphism (Mediator, Me)

Source: McLean, G., & Osei-Frimpong, K. (2019). Hey Alexa... examine the variables influencing the use of artificial intelligent in-home voice assistants. *Computers in Human Behavior*, 99, 28-37.

Items:

- When I communicate with the digital doctor, it feels like someone is near me
- The interaction experience with the digital doctor is close to that with a human being
- During interactions the with the digital doctor, I feel like I am communicating with a human
- I interact with the with the digital doctor in a way similar to how I interact with people

Intention to use (DV)

Source: Fishbein, M., & Ajzen, I. (1977). Belief, attitude, intention, and behavior: An introduction to theory and research. *Philosophy and Rhetoric*, 10(2), 177–188.

Items:

- I intend to use such digital-based healthcare service in the future
- Probably, I will use such digital-based healthcare service in the future

Sociodemographic variables

Sex: Male (He/Him) – Female (She/Her)

Age: [Year of birth]

Education: Lower than high school diploma; High School Diploma; B.Sc.; M.Sc.; Ph.D.

Employment: Employed; Unemployed; Unable to work; Retired; Student.

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